Probabilistic neural surrogates for uncertainty-to-risk analysis in coupled multi-physics systems: application to a heat-pipe-cooled reactor

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Abstract

We develop a probabilistic surrogate framework for coupled thermo-mechanical analysis of a heat-pipe-cooled reactor (HPR), enabling end-to-end uncertainty-to-risk quantification at a fraction of the cost of high-fidelity simulation. The surrogate is a heteroscedastic multilayer perceptron (HeteroMLP) trained on 2900 coupled OpenMC–FEniCS simulations, predicting distributions over 15 physics outputs from 8 material uncertainty inputs across two fidelity levels: a decoupled first-pass prediction (iter1) and a converged coupled steady-state (iter2). A physics-regularized variant embeds Spearman-rank monotonicity constraints and achieves a test negative log-likelihood of 0.305, with iter2 maximum global stress $\boldsymbol{R^2 = 0.929}$ (RMSE = 7.9 MPa). Forward propagation of 20 000 prior Monte Carlo samples reveals that multi-physics coupling substantially compresses uncertainty: the iter2 stress mean is 39 MPa lower than iter1 (153 MPa vs. 193 MPa), with standard deviation reduced by 47%; keff uncertainty collapses by **183**×. The predicted probability of exceeding the 131 MPa design threshold is **0.847** under the prior. Sobol sensitivity analysis identifies SS316 thermal conductivity (SS316_k_ref) as the dominant stress driver ($\boldsymbol{S_1 = 0.529}$) and thermal expansion coefficient (alpha_base) as the dominant keff driver ($\boldsymbol{S_1 = 0.481}$), indicating independent physical pathways. Posterior calibration over 20 benchmark cases demonstrates that, for all ten extreme-stress scenarios with observed stress $\geq$ 220 MPa, the posterior probability of exceeding 131 MPa converges to 1.0. The surrogate achieves a single-sample GPU latency of 1.7×10^{-7} s, representing a speedup of more than $\boldsymbol{23 \times 10^6}$ over the coupled solver.

Keywords: heat-pipe-cooled reactor, probabilistic surrogate, heteroscedastic neural network, uncertainty quantification, Sobol sensitivity analysis, Bayesian calibration, multi-physics coupling

1 Introduction

Heat-pipe-cooled reactors (HPRs) are an emerging class of compact microreactors in which the core comprises tightly coupled heat pipes, fuel rods, and monolith structures [**?**]. Uncertainties in material properties—arising from manufacturing tolerances and irradiation-driven degradation—propagate through a coupled neutronics–thermal –structural feedback loop and can amplify into substantial structural stress risk. Prior high-fidelity analysis of the nominal MegaPower design reports peak monolith stresses reaching 169 MPa under normal operating conditions [**?**], which exceeds the yield strength of the SS316 stainless-steel monolith at reactor operating temperatures (131 MPa); characterising how material-parameter uncertainty translates into exceedance probability is therefore a primary safety concern. Quantifying this risk requires sampling over the full 8-dimensional material parameter space, which is infeasible with repeated high-fidelity (HF) solver evaluations: each coupled OpenMC–FEniCS run takes approximately one hour [**? ?**].

Existing surrogate approaches face complementary limitations. Gaussian process (GP) regression and polynomial chaos expansion (PCE) become computationally prohibitive as the input dimension grows, and require additional regularity assumptions that may not hold across the full uncertainty range [**? ?**]. Physics-informed neural networks (PINNs) are designed for problems where exact governing equations can be expressed as residuals; the iterative coupling boundary conditions in an HPR preclude this formulation [**?**]. Data-driven neural surrogates offer scalable inference but typically lack uncertainty calibration and physical consistency guarantees.

This work makes four contributions. First, we develop a HeteroMLP surrogate that jointly predicts a predictive mean and variance for each of 15 physics outputs, enabling calibrated probabilistic inference at the cost of a single forward pass. Second, we propose a physics-regularized training objective that augments the Gaussian negative log-likelihood (NLL) with a Spearman-rank monotonicity term encoding known input–output relationships, and demonstrate substantially improved out-of-distribution (OOD) generalization. Third, we present a complete uncertainty-to-risk pipeline: forward Monte Carlo propagation, global Sobol sensitivity analysis with bootstrap confidence intervals, and MCMC-based posterior calibration. Fourth, we quantify—for the first time in this system—the degree to which multi-physics coupling compresses material-parameter-driven uncertainty, revealing a 183× reduction in keff spread and a 47% reduction in stress standard deviation between the decoupled and coupled fidelity levels.

2 Methods

2.1 Material uncertainty model

Eight material parameters are treated as independent uncertain inputs (Table **??**): the temperature-dependent Young's modulus $E(T) = E_{\text{slope}} \cdot T + E_{\text{intercept}}$ (slope and intercept separately), Poisson's ratio ν, thermal expansion coefficient $\alpha(T) = \alpha_{\text{base}} + \alpha_{\text{slope}} \cdot T$ (base and slope separately), and three SS316 stainless-steel thermal

properties (T_{ref}, k_{ref}, α_{SS316}). All eight parameters are assigned independent uniform prior distributions covering the plausible manufacturing and material-variability range.

Table 1 Input parameters and prior ranges.

Symbol	Description	Units	Prior type
E_{slope}	Young's modulus temp. slope	Pa/K	Uniform
$E_{\mathrm{intercept}}$	Young's modulus intercept	Pa	Uniform
ν	Poisson's ratio	—	Uniform
α_{base}	Thermal expansion (base)	K^{-1}	Uniform
α_{slope}	Thermal expansion (slope)	K^{-2}	Uniform
$T_{\mathrm{ref,SS316}}$	SS316 reference temperature	K	Uniform
$k_{\mathrm{ref,SS316}}$	SS316 thermal conductivity	W/(m·K)	Uniform
α_{SS316}	SS316 thermal expansion	K^{-1}	Uniform

2.2 Coupled high-fidelity simulation

The high-fidelity workflow couples OpenMC Monte Carlo neutronics [**?**] with the FEniCS finite-element solver [**?**] in an iterative feedback loop. Starting from initial material parameters, the neutronics calculation yields the effective neutron multiplication factor k_{eff} and the volumetric power distribution, which drives the FEniCS thermal–structural computation. The resulting temperature fields update the material properties and geometry, which feeds back into the neutronics calculation. Iteration continues until k_{eff} and the temperature field converge to a self-consistent steady state.

We define two fidelity levels: *iter1* (decoupled prediction)—the outputs of a single, uncoupled simulation pass without feedback; and *iter2* (coupled steady-state)—the outputs of the fully converged coupled iteration. Each evaluation takes approximately one hour on a dedicated compute node.

A dataset of $n = 2900$ samples was generated by Latin hypercube sampling of the 8-dimensional input space, covering the full prior uncertainty range. Each sample provides 15 physics outputs: 7 iter1 quantities (average fuel temperature, maximum fuel temperature, maximum monolith temperature, maximum global stress, monolith temperature, core height after expansion, wall expansion) and 8 iter2 quantities (k_{eff} plus the same 7 structural/thermal outputs). The dataset was split into training ($n_{\mathrm{train}} = 2029$), validation ($n_{\mathrm{val}} = 436$), and test ($n_{\mathrm{test}} = 435$) sets using a fixed random seed prior to any model selection.

2.3 Physics-regularized probabilistic surrogate

2.3.1 Network architecture

The surrogate is a HeteroMLP [**?**] that maps the 8-dimensional material parameter vector $\boldsymbol{\theta}$ to a predictive mean $\boldsymbol{\mu}(\boldsymbol{\theta})$ and log-variance $\log \boldsymbol{\sigma}^2(\boldsymbol{\theta})$ for each of the 15

physics outputs. At test time, the predictive distribution for output j is $p(y_j \mid \boldsymbol{\theta}) = \mathcal{N}(\mu_j(\boldsymbol{\theta}), \sigma_j^2(\boldsymbol{\theta}))$.

2.3.2 Training objectives

The *baseline surrogate* (Level 0) is trained by minimising the heteroscedastic Gaussian negative log-likelihood:

$$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{15} \left[\frac{(y_{ij} - \mu_{ij})^2}{2\,\sigma_{ij}^2} + \frac{1}{2} \log \sigma_{ij}^2 \right], \tag{1}$$

where $(y_{ij}, \boldsymbol{\theta}_i)$ are training samples.

The *physics-regularized surrogate* (Level 2) augments Eq. (**??**) with a monotonicity regularization term. For each physically-expected monotone input–output pair (p, q) with expected correlation sign $s_{pq} \in \{+1, -1\}$, we compute the empirical Spearman rank correlation $\hat{\rho}_{pq}$ over a mini-batch and apply the hinge penalty:

$$\mathcal{L}_{\mathrm{mono}} = \lambda \sum_{(p,q)} \max(0,\ \delta - s_{pq}\,\hat{\rho}_{pq})\,, \tag{2}$$

where λ is a regularization weight and δ is a margin hyperparameter. The total loss is $\mathcal{L} = \mathcal{L}_{\mathrm{NLL}} + \mathcal{L}_{\mathrm{mono}}$.

2.3.3 Hyperparameter optimisation

Hyperparameters for both Level 0 and Level 2 (number of hidden layers, hidden width, learning rate, batch size, λ, δ) were independently optimised using Optuna [**?**] with 40 trials per level, maximising validation NLL. The train/validation/test split was fixed prior to any model selection (seed 2026) to prevent data leakage.

2.4 Forward uncertainty propagation

20 000 samples are drawn from the joint prior distribution of the 8 input parameters. For each sample $\boldsymbol{\theta}_i$, the surrogate returns the predicted mean $\boldsymbol{\mu}(\boldsymbol{\theta}_i)$ for all 15 outputs. The distribution of these predicted means constitutes the epistemic (model-mean) forward uncertainty propagation. The primary quantity of interest is the second-iteration maximum global stress; the design threshold is $\tau = 131\,\mathrm{MPa}$—the yield strength of the SS316 stainless-steel monolith at reactor operating temperature [**?**] (main text; 110 MPa and 120 MPa thresholds in Appendix **??**). The failure probability is estimated as

$$P_{\mathrm{fail}}(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\mu_{\mathrm{stress}}(\boldsymbol{\theta}_i) > \tau]\,. \tag{3}$$

2.5 Global sensitivity analysis

First-order (S_1) and total-order (S_T) Sobol indices [**?** **?**] are estimated using Saltelli's quasi-random sampling scheme [**?**] with a base sample size of $N_S = 4096$ and 512 bootstrap replicates for 90% confidence intervals. The Level 2 surrogate mean prediction $\mu_{\text{stress}}(\boldsymbol{\theta})$ and $\mu_{k_{\text{eff}}}(\boldsymbol{\theta})$ serve as the function evaluators.

2.6 Observation-driven posterior calibration

For 20 benchmark test cases drawn from the held-out test set, MCMC [**?**] with a Gaussian likelihood is used to update the prior over four calibrated parameters ($E_{\text{intercept}}$, α_{base}, α_{slope}, ν), holding the remaining four inputs at their prior means. Observations consist of the true outputs corrupted by Gaussian noise. Each MCMC run generates 1200 posterior samples (acceptance rates 0.40–0.53). The posterior safety fraction is defined as

$$P_{\text{safe}}(\tau) = P(\mu_{\text{stress}}(\boldsymbol{\theta}) < \tau \mid \text{data}), \tag{4}$$

estimated from the posterior predictive samples.

Separately, 10 extreme-stress benchmark cases with true observed stress $\geq 220\,\text{MPa}$ are included to test posterior updating in safety-relevant extrapolation scenarios.

3 Results

3.1 Surrogate accuracy and physics regularization

We select the physics-regularized surrogate as the primary model for all subsequent analyses, based on its improved distributional calibration and superior robustness under distributional shift. The selected model achieves a test NLL of 0.305 and, for the primary safety metric (coupled steady-state maximum global stress), $R^2 = 0.929$, RMSE = 7.9 MPa, and 90% prediction interval coverage (PICP_{90}) of 0.913. Across the five primary outputs, the mean R^2 is 0.795 (Table **??**).

Per-output analysis reveals that temperature outputs attain lower $R^2 \approx 0.58$–0.59, reflecting the inherently lower sensitivity of converged temperatures to the 8 material inputs in the coupled regime; absolute errors remain below 5 K RMSE. Wall expansion achieves $R^2 = 0.996$. For k_{eff}, the surrogate reaches $R^2 = 0.874$.

As a supporting robustness check, we evaluate the surrogate on out-of-distribution inputs with extreme α_{base} values (Table **??**). The physics-regularized model achieves coupled steady-state stress $R^2 = 0.944$ under this extrapolation, compared to 0.774 for the baseline—a difference of $\Delta R^2 = +0.170$ that is consistent with the enforcement of the α_{base}–stress monotonicity relationship. A comparison of both models across all input features is provided in Appendix **??**.

Table 2 Per-output accuracy of the physics-regularized surrogate (Level 2) on the held-out test set ($n = 435$). Primary outputs are marked with ($\star$).

Output	Iter	MAE	RMSE	R^2	PICP_{90}	MPIW_{90}
k_{eff} ($\star$)	iter2	2.0×10^{-4}	2.5×10^{-4}	0.874	0.975	1.5×10^{-3}
Max. fuel temp. (K) ($\star$)	iter2	3.5	4.5	0.584	0.892	14.3
Max. monolith temp. (K) ($\star$)	iter2	4.0	5.1	0.589	0.899	16.2
Max. global stress (MPa) ($\star$)	iter2	6.1	7.9	0.929	0.913	28.6
Wall expansion (mm) ($\star$)	iter2	1.3×10^{-3}	1.6×10^{-3}	0.996	0.989	1.3×10^{-2}
Max. global stress (MPa)	iter1	8.8	11.3	0.928	0.920	42.6
Max. fuel temp. (K)	iter1	5.5	7.1	0.813	0.897	23.4

Source: `paper_metrics_per_dim_level2.csv`. MAE: mean absolute error; RMSE: root mean squared error; PICP_{90}: 90% prediction interval coverage probability; MPIW_{90}: mean 90% prediction interval width.

Table 3 Out-of-distribution (OOD) evaluation: iter2 stress R^2 for held-out samples with extreme values of each input feature. Level 2 shows the largest improvement over Level 0 for α_{base} extrapolation.

OOD feature	n_{OOD}	Level 0 R^2	Level 2 R^2
$E_{\mathrm{intercept}}$	577	0.940	0.949
α_{base}	580	0.774	**0.944**
ν	575	0.922	0.917
α_{slope}	576	0.909	0.934

Source: `paper_ood_multi_feature_summary.csv`.

3.2 Forward uncertainty propagation and threshold risk

Forward propagation of 20 000 prior samples through the Level 2 surrogate reveals that multi-physics coupling substantially reshapes the stress uncertainty distribution (Table **??**).

Multi-physics coupling substantially reshapes both the stress and k_{eff} uncertainty distributions (Table **??**, Fig. **??**).

In the decoupled prediction, the maximum global stress has a mean of 192.7 MPa and a standard deviation of 40.9 MPa (95th percentile: 264.8 MPa). The k_{eff} distribution is correspondingly broad, with a standard deviation of 0.0625 ($\approx$ 6250 pcm).

In the coupled steady-state, both distributions narrow markedly. The stress mean falls to 153.4 MPa—a reduction of 39.3 MPa (20%)—and the standard deviation decreases from 40.9 to 21.7 MPa (47% reduction). The compression of k_{eff} is more pronounced: the coupled steady-state standard deviation is 3.42×10^{-4} ($\approx$ 34 pcm), approximately 183$\times$ smaller than the decoupled value. The underlying mechanism is a negative feedback: materials with high thermal expansion coefficients expand the core

geometry under heating, which reduces neutron leakage and raises k_{eff}, in turn moderating local power density and partially offsetting the induced stress. This feedback is absent in the decoupled prediction.

The probability of exceeding the design threshold $\tau = 131\,\text{MPa}$ is estimated at $P_{\text{fail}} = 0.847$ under the prior (epistemic, mean predictions). The stress distribution has a long upper tail (maximum $\approx 229\,\text{MPa}$), consistent with the breadth of the material uncertainty range. This high exceedance probability is physically consistent with prior high-fidelity analysis of the nominal MegaPower design, which reports peak monolith stresses reaching 169 MPa under normal operating conditions [**?**]; the training dataset therefore predominantly samples the above-threshold operating regime, and the high P_{fail} reflects the position of the nominal design relative to the material yield limit rather than a surrogate artefact. Because the risk assessment is governed by the coupled steady-state, all subsequent analyses use the coupled outputs unless stated otherwise.

Table 4 Comparison of decoupled prediction (iter1) and coupled steady-state (iter2) uncertainty distributions for key outputs, from 20 000 prior Monte Carlo samples through the Level 2 surrogate (epistemic, mean predictions).

Output	iter1			iter2		
	Mean	Std	q_{95}	Mean	Std	q_{95}
k_{eff}	1.1025	0.0625	1.204	1.1035	3.4×10^{-4}	1.104
Max. fuel temp. (K)	1101.1	18.2	1135	1073.7	7.4	1088
Max. monolith temp. (K)	1067.0	22.4	1110	1035.4	7.6	1049
Max. global stress (MPa)	192.7	40.9	264.8	153.4	21.7	189.3
Wall expansion (mm)	29.920	0.0091	29.94	29.920	0.0082	29.93
$P_{\text{fail}}(\tau = 131\,\text{MPa}) = 0.847$ (iter2 stress, epistemic)						

Source: `forward_uq_all_outputs_mu_level2.csv`.

3.3 Global sensitivity analysis

Sobol sensitivity analysis identifies distinct dominant pathways for the two primary outputs of interest (Table **??**).

For iter2 maximum global stress (Level 2), the first-order index for SS316 thermal conductivity ($k_{\text{ref,SS316}}$) is $S_1 = 0.529$ (90% CI: 0.516–0.542), more than twice the next contributor. The Young's modulus intercept ($E_{\text{intercept}}$) ranks second with $S_1 = 0.202$ (CI: 0.180–0.225), followed by the SS316 reference temperature ($T_{\text{ref,SS316}}$) with $S_1 = 0.128$ (CI: 0.109–0.148). All three have well-separated, non-zero confidence intervals and together account for approximately 86% of the first-order variance decomposition. The remaining five inputs each have $S_1 < 0.07$, with confidence intervals that include or are near zero.

The dominance of $k_{\text{ref,SS316}}$ over stress reflects a direct physical link: the steel thermal conductivity governs how efficiently heat is transported from the fuel to the heat pipe, setting the local temperature gradient and the resulting thermal stress. High

conductivity reduces temperature gradients and mechanical stress; low conductivity concentrates heat and amplifies stress.

For iter2 k_{eff} (Level 2), α_{base} is the single dominant factor with $S_1 = 0.481$ (CI: 0.466–0.495), followed by α_{slope} ($S_1 = 0.065$, CI: 0.042–0.088). The remaining parameters have confidence intervals that include zero and are not treated as stable contributors. The dominance of α_{base} reflects that thermal expansion governs core geometry and thereby neutron leakage, the principal driver of k_{eff} variation.

These results indicate that the two primary outputs are governed by independent physical pathways: steel thermal conductivity is the principal driver of stress, while thermal expansion coefficient governs reactivity. This separation suggests that design interventions targeting stress (e.g., tighter control of $k_{\text{ref,SS316}}$) would not substantially perturb k_{eff} behaviour, and vice versa. The identification of α_{base} and α_{slope} as the dominant k_{eff} drivers, but not stress drivers, further motivates calibrating these parameters jointly with the elastic properties in the inverse analysis that follows.

Table 5 First-order Sobol indices S_1 (90% bootstrap CI) for iter2 maximum global stress and iter2 k_{eff} (Level 2 surrogate). Only factors with CI not including zero are listed; full results in Appendix **??**.

Output	Factor	S_1	CI low	CI high
Max. global stress	$k_{\text{ref,SS316}}$	0.529	0.516	0.542
	$E_{\text{intercept}}$	0.202	0.180	0.225
	$T_{\text{ref,SS316}}$	0.128	0.109	0.148
k_{eff}	α_{base}	0.481	0.466	0.495
	α_{slope}	0.065	0.042	0.088

Source: `paper_sobol_results_with_ci.csv`. Base sample $N_S = 4096$, 512 bootstrap replicates.

3.4 Observation-driven posterior calibration

MCMC posterior calibration was performed on 20 benchmark test cases with observed stress spanning approximately 100 to 226 MPa, as well as a separate set of 10 extreme-stress cases with observed stress ≥ 220 MPa. Acceptance rates across the 20 regular benchmark cases ranged from 0.40 to 0.53, indicating adequate MCMC mixing.

For cases with low observed stress (e.g., observed stress $\approx$ 102–114 MPa), the surrogate prediction at the posterior mean closely approximates the observation. For cases near the design threshold (≈ 131 MPa), the posterior mean prediction falls within a few MPa of the true value.

The posterior safety fraction $P_{\text{safe}}(131\,\text{MPa})$ (Eq. **??**) decreases systematically with increasing observed stress: cases with low observed stress yield $P_{\text{safe}} \approx 1.0$, while cases with high observed stress yield $P_{\text{safe}} \approx 0$.

The extreme-stress validation (Table **??**) demonstrates a consistent pattern: for all 10 cases with true stress ≥ 220 MPa, the posterior probability of exceeding 131 MPa

converges to $P_{\text{exceed}} = 1.0$, while the corresponding prior probability ranges from 0.84 to 0.96. The surrogate prediction at the posterior mean closely tracks the true stress in these cases (mean absolute error $< 6\,\text{MPa}$), consistent with the interpretation that the calibration localises the posterior near the correct high-risk parameter region. Sobol analysis (Section **??**) identifies low $k_{\text{ref,SS316}}$ as the dominant stress driver; the posterior systematically shifts toward this region following extreme-stress observations.

Table 6 Extreme-stress posterior risk updating: prior and posterior probabilities of exceeding 131 MPa for 10 benchmark cases with true stress $\geq 220\,\text{MPa}$.

True stress (MPa)	$P_{\text{exceed}}^{\text{prior}}$	$P_{\text{exceed}}^{\text{post}}$	Risk updated
278.4	0.943	1.000	Yes
261.4	0.939	1.000	Yes
253.0	0.926	1.000	Yes
241.3	0.957	1.000	Yes
240.1	0.951	1.000	Yes
237.9	0.885	1.000	Yes
236.0	0.934	1.000	Yes
233.0	0.841	≈1.000	Yes
225.7	0.818	≈1.000	Yes
220.5	0.911	≈1.000	Yes

Source: `paper_extreme_stress_risk_assessment.csv`.

4 Discussion

Multi-physics coupling as an uncertainty modulator.

The 47% reduction in stress standard deviation from iter1 to iter2 reflects a genuine negative feedback mechanism in the coupled physics, not a model artefact. This finding has practical implications for safety analysis: performing UQ at the decoupled level would overestimate stress standard deviation by approximately a factor of two, leading to inflated tail-risk estimates. The surrogate simultaneously models iter1 and iter2 outputs, enabling this comparison at negligible computational cost.

Physics regularization and model selection.

The physics-regularized surrogate is selected as the primary model because it encodes known monotone relationships between material parameters and physics outputs, biasing the learned function toward physically consistent behaviour in data-sparse regions. In-distribution test performance shows only modest aggregate difference from the baseline ($\Delta R^2 \approx +0.005$; see Appendix **??**). The more informative comparison is distributional: for α_{base} extrapolation the regularized model retains $R^2 = 0.944$ while the baseline falls to 0.774, a difference of +0.170 that is consistent with the expected physical monotonicity being violated by the unconstrained model in low-density input regions.

The discrepancy in Sobol-dominant factor between the baseline ($E_{\text{intercept}}$, $S_1 = 0.711$) and the regularized model ($k_{\text{ref,SS316}}$, $S_1 = 0.529$) is noted. The $k_{\text{ref,SS316}}$ attribution is more directly connected to the thermal stress mechanism; both results are reported in Appendix **??**.

Computational efficiency.

The surrogate achieves a single-sample CPU latency of $\approx 1.55 \times 10^{-4}$ s and a per-sample GPU batch latency of $\approx 1.70 \times 10^{-7}$ s, compared to ≈ 3600 s for the coupled HF solver. The resulting speedup is 23.2×10^6 (single CPU sample) or 21.2×10^9 (GPU batched), enabling full UQ analyses—including 20 000 Monte Carlo samples, Sobol analysis, and 20-case MCMC calibration—within minutes of wall-clock time.

Limitations.

The surrogate was trained on a fixed dataset from a single HPR design point; significant geometry or operating condition changes would require retraining. The OOD tests show that Level 2 degrades more gracefully than Level 0 in extrapolation, but absolute accuracy still decreases outside the training range. The MCMC calibration fixes four of the eight input parameters at prior means; a full 8-dimensional calibration would increase posterior uncertainty and may improve calibration for extreme scenarios. The posterior validation in Section **??** uses the existing HF dataset as a nearest-neighbour proxy rather than a true HF rerun at the posterior mean; the latter would provide a more direct validation but requires additional HF compute.

5 Conclusion

This work presents a probabilistic surrogate framework for coupled multi-physics uncertainty quantification in a heat-pipe-cooled reactor, demonstrating four principal findings:

1. **Physics regularization improves generalization.** The Level 2 HeteroMLP, trained with Spearman-rank monotonicity constraints, matches in-distribution accuracy (test NLL 0.305, stress $R^2 = 0.929$) while substantially improving OOD performance for α_{base} extrapolation (R^2: $0.774 \rightarrow 0.944$). This supports embedding domain knowledge as a soft regularizer.
2. **Multi-physics coupling compresses uncertainty.** Forward propagation shows that coupled steady-state outputs have significantly tighter distributions than decoupled predictions: 47% stress standard deviation reduction and 183× keff uncertainty compression. These are consequences of the neutronics–thermal feedback mechanism and cannot be captured by single-physics surrogates.
3. **Distinct sensitivity pathways exist.** Sobol analysis identifies $k_{\text{ref,SS316}}$ as the dominant stress driver ($S_1 = 0.529$) and α_{base} as the dominant keff driver ($S_1 = 0.481$), indicating that design interventions for stress reduction and reactivity control are largely independent.
4. **Posterior calibration sharpens risk assessments.** MCMC-based parameter updating conditioned on observations converges $P_{\text{exceed}}(131\,\text{MPa})$ to 1.0 for all

ten extreme-stress scenarios, demonstrating that the pipeline correctly identifies high-risk parameter regions from observational evidence.

The computational speedup ($> 23 \times 10^6$) makes this framework practical for iterative design exploration and real-time risk monitoring in reactor systems with significant material uncertainty.

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Declarations

- **Funding:** [TBD]
- **Conflict of interest:** The authors declare no competing interests.
- **Data availability:** The dataset and trained model checkpoints will be made available upon reasonable request.
- **Code availability:** Analysis code will be made available upon reasonable request.
- **Author contributions:** [TBD]

Appendix A Additional threshold analysis

Table **??** reports the posterior predictive failure probability at all three design thresholds for both Level 0 and Level 2.

Table A1 Predicted failure probability $P_{\text{fail}}(\tau)$ at three stress thresholds (epistemic, 20 000 samples).

Level	$P(\sigma > 110\,\text{MPa})$	$P(\sigma > 120\,\text{MPa})$	$P(\sigma > 131\,\text{MPa})$
Level 0	1.000	1.000	1.000
Level 2	0.979	0.938	0.847

Source: `paper_safety_threshold_summary.csv`. Level 0 values at 110 and 120 MPa exceed 1.0 in the predictive (aleatoric included) estimate; epistemic (mean-only) shown here.

Appendix B Full Sobol results

Table **??** provides first-order and total-order Sobol indices for all input–output pairs for both surrogate levels.

Appendix C Per-output accuracy table (all 15 outputs)

The full per-output accuracy metrics for the Level 2 surrogate on the test set are provided in Table **??**.

Table B2 First-order (S_1) Sobol indices for iter2 maximum global stress and iter2 k_{eff}, Level 0 and Level 2. Entries with CI including zero are marked with (†).

Output	Factor	S_1 (L0)	S_1 (L2)	L2 CI low	L2 CI high
Stress	E_{slope}	0.035	0.024	0.004	0.044
	$E_{\mathrm{intercept}}$	0.711	0.202	0.180	0.225
	ν	0.044	0.060	0.040	0.080
	α_{base}	0.090	0.052	0.028	0.076
	α_{slope}	$0.000^{\dagger}$	$0.011^{\dagger}$	−0.009	0.031
	$T_{\mathrm{ref,SS316}}$	$0.000^{\dagger}$	0.128	0.109	0.148
	$k_{\mathrm{ref,SS316}}$	$0.011^{\dagger}$	0.529	0.516	0.542
	α_{SS316}	$0.000^{\dagger}$	$0.017^{\dagger}$	−0.002	0.036
k_{eff}	E_{slope}	$0.002^{\dagger}$	$0.016^{\dagger}$	−0.008	0.040
	$E_{\mathrm{intercept}}$	0.020	$0.000^{\dagger}$	−0.029	0.021
	ν	0.038	$0.000^{\dagger}$	−0.035	0.016
	α_{base}	0.740	0.481	0.466	0.495
	α_{slope}	0.159	0.065	0.042	0.088
	$T_{\mathrm{ref,SS316}}$	$0.000^{\dagger}$	$0.000^{\dagger}$	−0.024	0.022
	$k_{\mathrm{ref,SS316}}$	$0.000^{\dagger}$	0.155	0.135	0.176
	α_{SS316}	$0.000^{\dagger}$	$0.000^{\dagger}$	−0.032	0.017

Source: `paper_sobol_results_with_ci.csv`.

Appendix D Speedup benchmark details

The computational speedup of the Level 2 surrogate versus the coupled HF solver is summarised in Table **??**.

Table C3 Complete per-output accuracy metrics, Level 2 surrogate, test set ($n = 435$). Primary outputs marked ($\star$).

Output	Iter	Group	MAE	RMSE	R^2	PICP_{90}
Avg. fuel temp. (K)	iter1	sec	3.08	4.02	0.820	0.894
Max. fuel temp. (K)	iter1	sec	5.49	7.09	0.813	0.897
Max. monolith temp. (K)	iter1	sec	6.21	8.04	0.819	0.885
Max. global stress (MPa)	iter1	sec	8.84	11.25	0.928	0.920
Monolith new temp. (K)	iter1	sec	1.04	1.36	0.823	0.901
Core height after (mm)	iter1	sec	0.017	0.021	0.987	1.000
Wall expansion (mm)	iter1	sec	1.2×10^{-3}	1.5×10^{-3}	0.996	0.995
k_{eff} ($\star$)	iter2	pri	2.0×10^{-4}	2.5×10^{-4}	0.874	0.975
Avg. fuel temp. (K)	iter2	sec	1.89	2.43	0.584	0.887
Max. fuel temp. (K) ($\star$)	iter2	pri	3.52	4.52	0.584	0.892
Max. monolith temp. (K) ($\star$)	iter2	pri	3.96	5.07	0.589	0.899
Max. global stress (MPa) ($\star$)	iter2	pri	6.10	7.91	0.929	0.913
Monolith new temp. (K)	iter2	sec	0.51	0.67	0.598	0.897
Core height after (mm)	iter2	sec	0.070	0.607	0.059	0.991
Wall expansion (mm) ($\star$)	iter2	pri	1.3×10^{-3}	1.6×10^{-3}	0.996	0.989

Source: `paper_metrics_per_dim_level2.csv`. pri: primary output; sec: secondary output.

Table D4 Computational speedup of the surrogate versus the HF solver.

Metric	Value	Notes
Single-sample CPU latency	1.55×10^{-4} s	Level 2, CPU
Per-sample GPU batch latency	1.70×10^{-7} s	Batch size 1024
HF solver (one case)	3600 s	OpenMC + FEniCS, 1 CPU node
Speedup (single CPU sample)	23.2×10^{6}	
Speedup (GPU batch per sample)	21.2×10^{9}	

Source: `paper_speedup_benchmark.json`.